Theory iFo: Focus on Likely Classes for Test-Time Prediction

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Abstract

Theoretical analysis showing that iFo can be seen as a noise reduction technique. It tends to improve under the assumption that shared features being non-contrastive (i.e., contributing positively to the most likely classes) are more robust.

1. Full Theoretical Motivation

Analysis Outline: We first prove for a simple model that iFo makes predictions rely more on shared, non-contrastive features, i.e., features that contribute to all top-k class logits positively. Then we investigate when iFo improves predictions. We formalize our assumption that shared, non-contrastive features are more robust, i.e., less prone to noise. We show that the assumption implies that iFo can be seen as a noise reduction technique. It always improves (loss of) predictions for non-contrastive shared features, i.e., if the top-k classes have positive coefficients for the shared feature. For contrastive features, it does not.

Model: We assume three output classes $C = \{y_0, y_1, y_2\}$ and focus classes $F = \{y_0, y_1\}$. We have four input features $X = \{x_0, x_1, x_2, x_3\}$.

The input features x_j can be seen as originating from a prior layer l with parameters c' processing activations z, i.e.,

$$x_i = l(z|c') = \sum_j c'_{j,i} z_j,$$

where z_j are fixed upstream activations for the current sample and $c'_{j,i}$ are the trainable upstream parameters. This exactly corresponds to a pre-activation representation, or equivalently to a post-nonlinearity activation in a locally linear region (e.g., an active ReLU unit). We further assume

$$\sum_j z_j^2 = 1.$$

To avoid a large number of symmetric case distinctions, we additionally consider a symmetric sample with

$$x_i = 1 \qquad \text{for all } i.$$

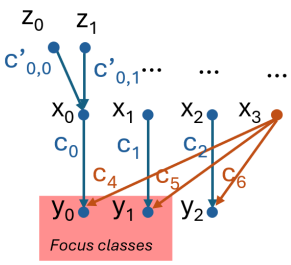

Figure 1. Theoretical model: 3 output classes y_i and one shared feature x_3 among all y_i

Outputs $y_i := f_i(X)$ are computed as:

$$\begin{aligned} y_0 &= c_0 x_0 + c_4 x_3 \\ y_1 &= c_1 x_1 + c_5 x_3 \\ y_2 &= c_2 x_2 + c_6 x_3 \end{aligned} \tag{1}$$

We apply the intuitive notion that a feature value cannot be negative, i.e., $x_i \geq 0$, which happens, for example, after layer activations pass through a non-linearity like $ReLU$. We also assume that $c_i \neq 0$ to avoid edge cases that can easily be deduced from our analysis. The model implicitly expresses that the presence of input features $\{x_0, x_1, x_2\}$ is only indicative of class i, i.e., a change of x_i alters only y_i for $i \in \{0, 1, 2\}$. In turn, $c_0 > 0$, $c_1 > 0$, and $c_2 > 0$. The shared feature x_3 impacts all outputs y_i. It might also be contrastive, i.e., it can be that two parameters from $\{c_4, c_5, c_6\}$ have opposing signs. Depending on the sign of c_4, c_5 and c_6 the presence of the feature ($x_3 > 0$) increases or decreases y_i. As we model dependence among classes only through the shared feature x_3, we assume that the upstream parameter blocks $c'_{\cdot,i}$ are feature-specific, so that updating the parameters producing x_i does not directly affect x_m for $m \neq i$. This also facilitates our analysis.

Thus, we do not show any preference for a shared or non-shared feature and its upstream features z_i, we discuss deviating from this assumption after the proofs as they clarify the behavior. We are mostly concerned with one layer being given by the coefficients c_i. We consider updates to parameters c_i depending on the loss from our methods iFo $L^{iFo} = -(y_0 + y_1)$ without weighting and from optimizing just a single class decreasing $L_{y_i,-} = -y_i$.

For an illustration of our model, see Figure 1.

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Preliminaries: *Let us compute updates to c_i. We need the following partial derivatives:*

$$\frac{\partial L_{y_j,-}}{\partial c_i} = \begin{cases} -x_j, & \text{if } i = j, \\ -x_3, & \text{if } i = 4+j, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

$$\frac{\partial L^{iFo}}{\partial c_i} = \begin{cases} -x_0, & \text{if } i = 0, \\ -x_1, & \text{if } i = 1, \\ -x_3, & \text{if } i \in \{4,5\}, \\ 0, & \text{otherwise}. \end{cases} \tag{3}$$

Note that as $x_i = 1$, all non-zero updates to c_i are identical in magnitude.

To study changes in the features, we consider one gradient step on the upstream parameters $c'_{j,i}$. Since

$$x_i = \sum_j c'_{j,i} z_j,$$

we have

$$\frac{\partial L}{\partial c'_{j,i}} = \frac{\partial L}{\partial x_i}\frac{\partial x_i}{\partial c'_{j,i}} = \frac{\partial L}{\partial x_i} z_j.$$

Hence, for learning rate $\lambda = 1$,

$$\Delta x_i = \sum_j \Delta c'_{j,i} z_j = -\lambda \sum_j \frac{\partial L}{\partial c'_{j,i}} z_j \tag{4}$$

$$= -\frac{\partial L}{\partial x_i} \sum_j z_j^2 \tag{5}$$

$$= -\frac{\partial L}{\partial x_i}. \tag{6}$$

The feature-level derivatives are therefore

$$\frac{\partial L_{y_j,-}}{\partial x_i} = \begin{cases} -c_j, & i = j, \\ -c_{4+j}, & i = 3, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

$$\frac{\partial L^{iFo}}{\partial x_i} = \begin{cases} -c_0, & i = 0, \\ -c_1, & i = 1, \\ -(c_4 + c_5), & i = 3, \\ 0, & \text{otherwise}. \end{cases} \tag{8}$$

Thus, for $\lambda = 1$, the induced feature changes are

$$\Delta x_i^{y_j} = \begin{cases} c_j, & i = j, \\ c_{4+j}, & i = 3, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

$$\Delta x_i^{iFo} = \begin{cases} c_0, & i = 0, \\ c_1, & i = 1, \\ c_4 + c_5, & i = 3, \\ 0, & \text{otherwise}. \end{cases} \tag{10}$$

Theorem 1.1. *For a non-contrastive shared feature (that is, $\operatorname{sgn}(c_4) = \operatorname{sgn}(c_5)$), updates to a shared feature x_3 are larger in magnitude for iFo ($k = 2$) than for the most likely class($k = 1$). For contrasting features, updates are smaller in magnitude for iFo than when using only the most likely class. For a non-shared feature of the most likely class y_i updates to x_i are identical.*

Proof. For the shared feature x_3 for iFo we have that its change is $\Delta x_3 = c_4 + c_5$ and for the most likely class either proportional to c_4 or c_5 depending on whether y_0 or y_1 is the most likely class. If a feature is non-contrastive then we have $\operatorname{sgn}(c_4) = \operatorname{sgn}(c_5)$ and thus $|c_4 + c_5| > \max(|c_4|, |c_5|)$, i.e., the update is larger for non-contrastive shared features for iFo ($k = 2$) and when updating the most likely class $k = 1$ (which can be either y_0 or y_1).

If $\operatorname{sgn}(c_4) \neq \operatorname{sgn}(c_5)$, then $|c_4 + c_5| < \max(|c_4|, |c_5|)$ and thus the update is smaller. For non-shared features x_0, x_1 the update is $\Delta x_j = c_j$ for $j \in \{0, 1\}$. □

Next, we investigate when iFo improves predictions (and when it does not).

Assume all parameters originating from training are perfect, so that without noise the predictions are always correct. At test time, we assume the presence of noise, e.g., to account for differences between training and test data distribution. Let us assume that non-contrastive shared features are more robust, i.e. less noisy, i.e., noise on x_3 is zero. Wlog, assume only x_0 is noisy (the case for x_1 is analogous), i.e.

$$\hat{x}_0 = x_0 + R_0, \qquad R_0 \sim \text{Uniform}[-1, 1].$$

Also, coefficients $c_0 = c_2 = c_5 = 1$. The coefficient c_4 for the shared feature is $c_4 = 1 + \delta_4$ and for the non-shared feature x_1 we have $c_1 = 1 + \delta_1$. Then

$$\hat{y}_0 = y_0 + c_0 R_0 = y_0 + R_0.$$

As we do not constrain δ_1, δ_4 to be both either positive or negative, this setup covers many cases. It is also motivated by the fact that these assumptions lead to similar outputs y_i if $|\delta_4|$ and $|\delta_1|$ are small. In the most extreme case, when $\delta_4 = \delta_1 = 1$ we get maximal uncertainty without noise $R_0 = 0$, meaning $y_0 = y_1$. To avoid this edge case, we assume $\delta_1 \neq \delta_4$.

Note, that a key quantity in our analysis is $c_4 + c_5 = (1 + \delta_4) + 1$. We call a feature positively shared if $\delta_4 > -1$ so that $c_4 > 0$ and $c_5 > 0$ and, thus, $|c_4 + c_5| > \max(|c_4|, |c_5|)$. Wlog. assume a learning rate of $\lambda = 1$.

Let us focus on the situation, where the correct class is within the top-k classes and thus only analyze the outputs y_0, y_1. This is the most interesting case, as otherwise the

prediction on the original classifier is already wrong (and thus changing the prediction in any way cannot cause a wrong prediction as it is already wrong, i.e., iFo might only help).

Theorem 1.2. *If the feature x_3 is positively shared, i.e.$\delta_4 > -1$ and other features have identical coefficients $c_i = 1$ then iFo improves the margin between the correct and wrong class. More generally, if $\delta_4+\delta_1 > -2$ then iFo improves the margin between the correct and wrong class. Otherwise, iFo does not improve the margin; in particular, if $\delta_4 + \delta_1 < -2$, it changes the margin in favor of the wrong class.*

Let us discuss the theorem less formally. The first sentence says that iFo helps if a feature is shared positively and non-shared features have identical coefficients. The second sentence generalizes this statement removing the requirement that non-shared features need identical coefficients. It shows that if a feature is shared positively, the non-shared feature c_1 can have any coefficient $c_1 > 0$, i.e., $\delta_1 > -1$. In fact, this is no restriction at all following our notion that a feature x_i is indicative of one class and and $x_i > 0$ (e.g., due to ReLu) and thus $c_i > 0$.

Proof. For the unmodified model, the gap between the two most likely classes is

$$y_0 - y_1 = R_0 + \delta_4 - \delta_1.$$

For iFo we perform one gradient step with learning rate 1. Then

$$\Delta x_3 = c_4 + c_5, \qquad \Delta x_0 = c_0, \qquad \Delta x_1 = c_1.$$

The resulting logit updates are

$$\Delta y_0' = c_0^2 + c_4(c_4 + c_5), \qquad \Delta y_1' = c_1^2 + c_5(c_4 + c_5).$$

Substituting the c_i's,

$$\Delta y_0 = 3 + 3\delta_4 + \delta_4^2, \qquad \Delta y_1 = 3 + \delta_4 + 2\delta_1 + \delta_1^2,$$

and hence, the change in the gap of y_0 and y_1 due to iFo is:

$$\Delta y_0 - \Delta y_1 = 2\delta_4 + \delta_4^2 - 2\delta_1 - \delta_1^2.$$

After iFo, the gap becomes

$$\begin{aligned} &(y_0 + \Delta y_0) - (y_1 + \Delta y_1) && (11)\\ &= (y_0 - y_1) + (\Delta y_0 - \Delta y_1) && (12)\\ &= (R_0 + \delta_4 - \delta_1) + (2\delta_4 + \delta_4^2 - 2\delta_1 - \delta_1^2) && (13)\\ &= R_0 + 3\delta_4 + \delta_4^2 - 3\delta_1 - \delta_1^2 && (14) \end{aligned}$$

Case 1: $\delta_4 > 0$, $\delta_1 = 0$

Before iFo,

$$y_0 - y_1 = R_0 + \delta_4.$$

Thus, $y_0 > y_1$ without noise ($R_0 = 0$) as $\delta_4 > 0$ by assumption and y_0 is the correct class (as we assume parameters c_i are perfect), and the prediction of the unmodified model is correct whenever

$$R_0 > -\delta_4.$$

After iFo,

$$(y_0 + \Delta y_0) - (y_1 + \Delta y_1) = R_0 + 3\delta_4 + \delta_4^2.$$

If the prediction is already correct ($R_0 > -\delta_4$), then it remains correct for a positively shared feature as $3\delta_4 + \delta_4^2 \geq 0$ as by assumption $\delta_4 > -2$. For other features, it remains correct only if $\delta_4 > -3$.

If it is wrong ($R_0 < -\delta_4$), iFo improves the loss as $2\delta_4 + \delta_4^2 > 0$ and fixes the prediction if $R_0 + 3\delta_4 + \delta_4^2 > 0$. For positively shared features, we always improve as by assumption $\delta_4 > -2$ and we need $\delta_4 > -3$ for $3\delta_4+\delta_4^2 > 0$. Otherwise it fails for $\delta_4 < -2$.

Case 2: $\delta_4 < 0$, $\delta_1 = 0$

Before iFo,

$$y_0 - y_1 = R_0 + \delta_4.$$

Thus, $y_0 < y_1$ without noise ($R_0 = 0$) as $\delta_4 < 0$ by assumption and y_1 is the correct class, and the prediction is correct whenever

$$R_0 < -\delta_4.$$

After iFo,

$$(y_0 + \Delta y_0) - (y_1 + \Delta y_1) = R_0 + 3\delta_4 + \delta_4^2.$$

If the prediction is already correct ($R_0 < -\delta_4$), then it remains correct for equally shared features as $R_0 + 3\delta_4 + \delta_4^2 \leq 2\delta_4 + \delta_4^2 = \delta_4(2 + \delta_4) < 0$. This holds if $\delta_4 > -2$, which is true by assumption on δ_4 for a positively shared features and as $\delta_4 < 0$

If it is wrong ($R_0 > -\delta_4$), iFo improves the loss if $2\delta_4 + \delta_4^2 < 0$, which holds true as by assumption for equally shared features $|\delta_4| < 2$ and $\delta_4 < 0$. If $\delta_4 < -2$, iFo worsens the loss.

Case 3: $\delta_4 + \delta_1 > -2$

We factor the change in the gap as

$$\big[(y_0+\Delta y_0)-(y_1+\Delta y_1)\big]-(y_0-y_1) = (\delta_4-\delta_1)(2+\delta_4+\delta_1).$$

Since $\delta_4+\delta_1 > -2$, we have $2+\delta_4+\delta_1 > 0$, and therefore the change in the gap has the same sign as $\delta_4 - \delta_1$.

If $\delta_4 > \delta_1$, then without noise y_0 is the correct class, because

$$y_0 - y_1 = R_0 + \delta_4 - \delta_1$$

is positive at $R_0 = 0$. In this case,

$$(\delta_4 - \delta_1)(2 + \delta_4 + \delta_1) > 0,$$

so iFo increases the gap and thus improves the margin in favor of the correct class.

If $\delta_4 < \delta_1$, then y_1 is the correct class. In that case,

$$(\delta_4 - \delta_1)(2 + \delta_4 + \delta_1) < 0,$$

so iFo decreases $y_0 - y_1$, i.e. makes the gap more negative, again increasing the margin in favor of the correct class.

Equivalently, after iFo,

$$(y_0 + \Delta y_0) - (y_1 + \Delta y_1) = R_0 + (\delta_4 - \delta_1)(3 + \delta_4 + \delta_1).$$

Because $\delta_4 + \delta_1 > -2$ implies $3 + \delta_4 + \delta_1 > 1$, the magnitude of the noiseless margin is amplified. Hence iFo improves the margin whenever $\delta_4 + \delta_1 > -2$.

Case 4: $\delta_4 + \delta_1 < -2$

Now $2 + \delta_4 + \delta_1 < 0$, hence

$$(\delta_4 - \delta_1)(2 + \delta_4 + \delta_1)$$

has the opposite sign of $\delta_4 - \delta_1$. Therefore iFo always changes the gap in the direction of the wrong class.

If $\delta_4 > \delta_1$, then y_0 is the correct class in the noiseless model, but

$$(\delta_4 - \delta_1)(2 + \delta_4 + \delta_1) < 0,$$

so iFo reduces the gap $y_0 - y_1$ and thus shrinks the margin.

If $\delta_4 < \delta_1$, then y_1 is the correct class, but

$$(\delta_4 - \delta_1)(2 + \delta_4 + \delta_1) > 0,$$

so iFo increases $y_0 - y_1$, making it less negative and again shrinking the margin of the correct class.

Moreover,

$$(y_0 + \Delta y_0) - (y_1 + \Delta y_1) = R_0 + (\delta_4 - \delta_1)(3 + \delta_4 + \delta_1).$$

If $-3 < \delta_4 + \delta_1 < -2$, then $0 < 3 + \delta_4 + \delta_1 < 1$, so iFo preserves the ordering in the noiseless case but weakens the margin. If $\delta_4 + \delta_1 < -3$, then $3 + \delta_4 + \delta_1 < 0$, so iFo can even reverse the ordering in the noiseless case. Thus iFo does not improve in this regime.

□

Remark 1: For perfect symmetry $\delta_4 = \delta_1 = 0$, iFo has no impact, i.e.,

Before iFo,

$$y_0 - y_1 = R_0.$$

After iFo,

$$(y_0 + \Delta y_0) - (y_1 + \Delta y_1) = R_0.$$

Thus, the ground truth assigns equal likelihood to both classes, and the outcome depends only on noise. In this case, iFo has no effect.

Remark 2: Deviating from a symmetric setupa and preferring shared features $x_3 > x_i$ or updates to shared features is favorable for iFo and extends the range of δ_4 for which iFo is beneficial. In turn, preferring non-shared features, reduces the range of δ_4.